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Real Time Malicious Drone Detection Using Deep Learning on FANETs

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***Abstract*—Unmanned aerial vehicles, especially drones, are increasingly used for transportation, communication and military purposes. Using only one drone to accomplish a mission leads to a solution that is costly and has low error tolerance. For this reason, a network structure called Flying Ad-Hoc Networks (FANET) organize drones with a task sharing mechanism to lower cost. However, these networks remain vulnerable to various attacks due to vulnerabilities such as the use of civilian drones, use of unencrypted GPS signals, physical attacks using malicious drones, etc. Solutions to these attacks should consider the limited memory and calculation capabilities of drones. In this study, drone detection and subsequent classification of malicious drones, which are potential sources of man-in-the-middle or physical attacks, were implemented based on real images that can be captured by the drone camera. You Only Look Once (YOLO) detection algorithm was used in the drone detection phase and a Convolutional Neural Network model was used in the classification phase. In the study, a dataset consisting of 4 classes (Yuneec Typhon, DJI Tello, DJI Phantom 4 and Other) was created using internet resources and YouTube videos, and the classification success was measured as 88.78%.**

***Index Terms*—Computer Vision, Object Detection, Deep learning, CNN, image processing, FANET, Drone Networks.**

I. Introduction

The difficulty and high cost of creating physical network infrastructure for scenarios such as military operations and natural disasters leads to the use of ad-hoc networks where nodes are configured and coordinated instantly rather than using a pre-planned infrastructure. A flying ad-hoc network (FANET) is a group of drones that create a temporary network structure to achieve a civil or military purpose [1]. FANETs can be used for various purposes such as environmental imaging, military border security, disaster management, cargo transmission, and mapping [2]. The use of many small drones instead of one drone can create cost-effective solutions. With the use of multiple drones, errors occurring on a single drone can be tolerated by other drones and the network can be scaled according to the increasing or decreasing number of drones or the problem [3].

In FANETs, Global Positioning System (GPS) is generally used for positioning the drones in the network and movement of the members of the network is achieved by sending location information by means of Drone to Drone communications which is mainly achieved with wireless infrastructures [3]. Drone network communication is examined under two separate headings, which are communication of drones with each other and communication of drones with an infrastructure. The communication between drones can be seen as a peer-to-peer (P2P) network structure. The communication of drones with an infrastructure can be examined under 3 different categories:

- Drone-Satelitte Communication: It is used to communicate with the drone's GPS system. In GPS deception attacks, the aim is to direct the drone to a desired location by sending fake satellite signals to the drone as if they are coming from the satellite.
- Drone-Network Communication: It is useful for cellular networks and ensuring network security is critical.
- Drone-Ground Control Station Communication: It is based on wireless network technologies such as Bluetooth and Wi-Fi. Wireless network structures are accessible to everyone makes these communication infrastructures vulnerable to attacks.

FANETs generally consist of multiple drone devices, a ground station and wireless communication infrastructures to ensure communication [4]. With the increasing use of drones, various security problems have emerged due to their use of publicly available wireless communication infrastructures [5].

Drone systems may contain sensors such as cameras, temperature, IMUs, altimeters and barometers. Drone manufacturers do not encrypt sensor data generally and

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so attacks made on sensors by attackers can change temperature values, camera images or disable sensors. However, generally, the attacker must be close to the network to perform these attacks. That is why, it is important to detect malicious drones which is close to drone network and take necessary precautions to preserve whole network [3]. Moreover, physical attacks against drones are generally carried out with another drone which tries to get close to the network. Drone developed by the Delft Dynamics company allows another drone to be neutralized by throwing a parachute net in the environment and to drop it to the ground without being damaged [2].

Our contributions of the paper can be listed below.

- A CNN and YOLO based malicious drone detection and classification system is proposed to detect anomalous situation by using on board drone camera without human intervention.
- Proposed model is designed to be on board, low-cost and portable which allows quick detection of malicious situations on the platform and makes the communication encrypted only algorithm decides a suspicious situations.
- Model has faster recognition capabilities and only encrypted the communication when it recognizes a suspicious situation which extends battery life when compared the other solutions which are placed in literature.

II. Related Work

Machine learning-based approaches offer effective solutions against attacks such as detecting the attacker drone in the network, Man-in-the-Middle attacks, Denial of Service (DoS) attacks and GPS spoofing as a result of being able to process multidimensional data. Various algorithms such as CNN, SVM (Support Vector Machines), ANN (Artifical Neural Networks), RNN (Recurrent Neural Networks) are widely used in the literature to ensure drone security [6].

Sedjelmaci et al. [7], designed a machine learning-based detection system where any event was detected by a drone and a detection rate of over 93% was achieved. However, the model can be applied on a single drone, and it is recommended to apply the model on drone swarms within the scope of future studies. In the study of Wyder et al. [8], the target drone and its size were determined by the object recognition algorithm using the YOLO detection algorithm against GPS Deception and DoS attacks. 77% real-time classification success was achieved at 5.22 fps.

Chaozheng et al. [9] aimed to determine the drone class through IQ, FFT (Fast Fourier Transform) and spectrogram data of different drones. It is succesfully classified the different drone types but, requires the use of an SDR (Software Defined Radio) to record different drone's signatures. Since the operational range of acoustic systems is 200-250 meters, radio frequency based solutions are very useful to detect drones from longer ranges. Xue et al. [10] created a model which uses images taken from the drone camera were compared with satellite images and it was decided whether there was an attack or not according to a certain threshold value using the Euclidean distance. Study has the advantages of not requiring complex GPS deception and also can be produced on the drone.

Seidaliyeva et al. [11] performed real-time drone detection and drone presence classification through video images taken from a drone. Moving background data caused the classification time to extend. Speeding up the detection phase by using motion and visual data from the sensors together was proposed as a future step. Piczak et al. [12] classify drones based on drone sounds successfully. However, since the training set includes a small number of different drone class, diversification from different classes of drones should be considered.

In a study utilizing blockchain, Drone ID values, flight routes, flight programs, and sensor data were stored in the blockchain, ensuring that drones move at a safe distance from each other [13]. Drones are resource-constrained devices. Implementing security algorithms like blockchain will increase computation and storage costs, thereby increasing energy consumption and reducing flight time. Additionally, the delay caused by blockchain algorithms will pose a problem for drone applications requiring critical communication needs (such as location data).

III. Methodology

The methodology in this work consists of two stages as explained in the following subsections. The entire workflow is also demonstrated in Fig. 1.

A. YOLO Based Drone Detection

YOLO is a detection algorithm used to distinguish defined objects by processing an image or video in one step, identifying both the location and class of every object. In contrast to other object detection methods like RCNN, YOLO handles the image all at once instead of partitioning it into regions that results quicker processing time. This characteristic renders it appropriate for applications requiring real-time processing [14], [15]. It

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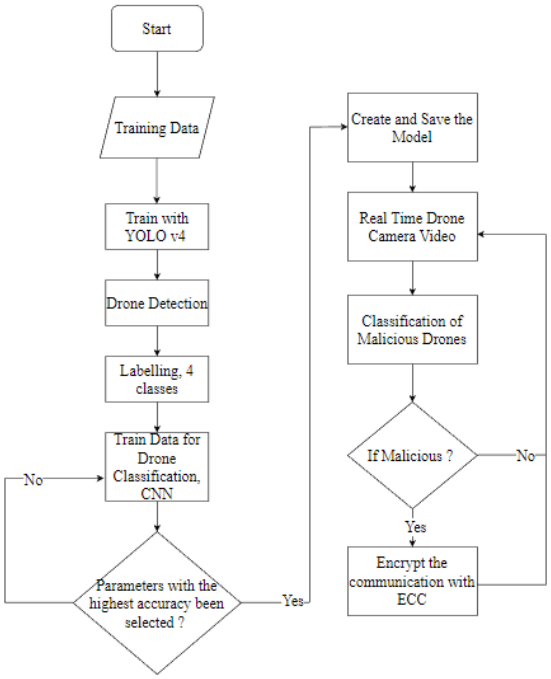


Fig. 1. The General Structure of Proposed Model

could be used in real-time drone identification in video footage, thereby playing a crucial role to be proactive for drone-related incidents [16].

In the drone classification process, initially detecting drones from the relevant image by using YOLO and then classifying them using the corresponding cropped image which is detected by the algorithm enhances performance. In this work, the YOLO algorithm was used for drone detection, while CNN was used for drone classification. During the training phase for drone detection, Google Colab was used because of Google Colab's provision of Tesla GPU hardware and its ability to obtain algorithm outputs faster [17].

Initially, images of different drones were obtained using Google Image Search. Labeling and object marking processes were carried out on the obtained images using the makesense.ai program. With the completion of the labeling process, training with YOLO was performed on Google Colab, and as a result of the training, config and weights files were obtained. With the creation of these files, an infrastructure capable of detecting drones from images was provided. The obtained config and weight files were saved and applied to a dataset created from approximately 500 images of DJI Tello, DJI Phantom4, Yuneec Typhoon, and Other drone classes obtained by ourselves from Google Image Search and YouTube videos. Thus, in the 4-class dataset created for drone classification, the elimination of other objects is ensured by obtaining the drone in the photo after drone detection. As a result of this preprocessing, a dataset prepared for the classifier to enter the drone classification algorithm is obtained after drone detection.

B. Convolutional Neural Networks

CNN algorithm can process images as input and extract distinctive features from them. It aims to identify patterns within images by scrutinizing characteristics such as color variations, saturation levels, and contrasts between light and dark tones [18]. CNN offers a wider scope of high-level feature extraction capabilities, resulting in superior performance in image and speech classification when contrasted with other deep learning algorithms [19], [20]. Convolutional layers in CNN is used to extract unique input features from data. These layers constitute essential components of CNNs, tasked with comprehending input characteristics through learnable filters. Although these filters are small in size, they cover the entire data like image and facilitate the transfer of learned features to subsequent layers. Subsequent of these layers are pooling layers, which are employed to decrease computational and memory demands. That is very important for drone based and also real-time calculation operations [20], [21]. While this operation serves computational and memory constraints well, it may result in the loss of less significant but still distinctive elements within the inputs [22]. Towards the conclusion of the CNN architecture, in fully-connected layers, weight values gathered from preceding layers are aggregated, and the input is flattened to compute the probability of each class encompassed within the algorithm [23].

In the classification phase of his work, CNN is trained for different combinations of parameters such as different filter sizes, neuron numbers, optimization functions, and layer numbers. Detailed training results are presented in the following sections. After the training is completed, the model is saved, and during the testing phase, the saved model is moved onto the real-time camera image obtained from the DJI Tello Drone, and real-time classification is performed. After detection, conceptually, drones in the drone network can share information such as speed, location, and battery level by encrypting them with ECC. Additionally, if the detected drone is a DJI Tello, sending the "command" command in the Tello SDK will cause the drone to detect the controller if it is not encrypted (it is not encrypted by default). The

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vulnerability here is the detection of any drone controller while connected, as the drone can perceive the attacker as the controller. This way, the attacker can take control of the drone, cut the camera connection of the real controller with the ”streamon” command, and directly drop the drone with the ”emergency” command [24].

C. DJI Tello Drone

DJI Tello drone connects the controller via Wi-Fi UDP (User Datagram Protocol) port and allows users to control drone with text commands.To receive Tello's video stream 192.168.10.1 IP address is used and the UDP server is set to IP address 0.0.0.0 and UDP port 1111. After that, sending ”streamon” command to Tello via UDP port 8889 starts streaming [24]. Properties of the DJI Tello is given Table I [25].

TABLE I. DJI Tello Properties

Feature	Value
Geolocalization	No GPS
Wi-Fi	IEEE 802.11n 2.4 GHz
Camera FOV	82.6○
Video resolution	1280 × 720 pixels (HD, 30 fps)
Cameras	Front-facing camera

IV. Experimental Evaluation

A. Dataset and Experimental Environment

We assembled a dataset containing approximately 500 images, encompassing DJI Tello, DJI Phantom4, Yuneec Typhoon, and Other drone classes which is shown in Figure 2. These images were gathered from Google Image Search and screenshot of YouTube videos. During the creation of a 4-class dataset for drone classification, we eliminated any irrelevant objects by isolating the drone in each photo after the drone detection process. Detection of drone from image with YOLO step resulted in a refined dataset, tailored for input into the drone classification algorithm.



Fig. 2. Dataset

Following the training of the CNN, the model's output is transferred to the Raspberry Pi and also deployed on the test computer, which processes real-time drone camera videos. Ubuntu operating system was installed on the Raspberry Pi that enables the implementation of the algorithm independently of the platform. As a result, a solution has been developed for both on the drone with the Raspberry Pi and on the ground control station with the test computer.

B. Algorithm Timing Performance

Within this section, real time performance of the detection and classification operation for the algorithm is evaluated. Owing to the fact that alternative hardware was not available, timing performance assessments of the algorithm across various platforms were not carried out. Deploying the algorithm on another hardware setups may lead to variations in results.

C. Performance Results and Discussion

In this section, all tests have been run 5 times, and the average values are provided in given tables. Performance of different optimization functions observed on the dataset and outcomes are given with Table II. It is seen that Adam (Adaptive Momentum) optimizer shows the best performance when compared to other optimization algorithms.

TABLE II. Results of Different Optimization Functions

Opt. Func.	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. F1-score
Ada Grad	86.91%	0.86	0.88	0.87
Adam	87.85%	0.87	0.89	0.88
SGD	87.85%	0.87	0.89	0.88
RMS Prop	78.50%	0.79	0.80	0.79

Table III displays the performance of different pooling algorithms, with max pooling being employed as the default in CNN and exhibiting superior performance on the dataset when compared to average pooling.

TABLE III. Results of Different Pooling Functions

Pooling Fun.	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. F1-score
Max Pool.	79.43%	0.79	0.83	0.80
Avg Pool.	85.04%	0.85	0.86	0.85

Using maximum pooling has been distinctive, but it has caused the loss of some distinguishing but not crucial features, leading to a decrease in performance.

The next experiment was conducted to observe the impact of using different filter sizes within the convolutional layers. In the existing literature, filter sizes of 3×3, 5×5, and 7×7 are commonly utilized. In addition to these filters, the effect of non-square dimensions of images resulting from detection with YOLO and the use of rectangular filters on the algorithm performance has also been investigated. Results are shown in Table IV.

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TABLE IV. Results of Different Filter Size

Filter Size	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. F1-score
3x3, 3x3, 3x3	88.78%	0.90	0.89	0.89
5x5, 5x5, 5x5	81.30%	0.81	0.84	0.82
7x7, 7x7, 7x7	85.04%	0.83	0.85	0.84
2x2, 2x2, 2x2	80.37%	0.82	0.81	0.81
5x3, 5x3, 5x3	81.30%	0.82	0.82	0.82
7x3, 7x3, 7x3	78.50%	0.79	0.79	0.79
3x5, 3x5, 3x5	80.37%	0.79	0.81	0.80
3x3, 5x5, 7x7	85.04%	0.84	0.87	0.85
7x7, 5x5, 3x3	84.11%	0.85	0.85	0.85

A filter used to highlight edges in an image can be 3x3 in size. This means that the filter matrix consists of 3 rows and 3 columns. Each element represents a specific weight, and the filter helps detect various features by sliding over the input image. Using small-sized filters will help transfer more features to the subsequent layers and improve performance. This can also be seen in Table IV. A 2x2 filter, on the other hand, has a lower performance value compared to 3x3 because it cannot consider all neighbors of the pixel. Additionally, the performance of non-square filters has also been examined.

The number of nodes across various convolutional layers is a significant parameter which affects the algorithm performance, and its impact has been examined within the study's scope in Table V

TABLE V. Effect of Different Denses in Convolutional Layers

Model	Kernel Size	Avg. Accuracy	Avg. Precision	Avg. Recall	Avg. F1-score
CNN	16, 16, 16	74.76%	0.73	0.75	0.74
CNN	64, 64, 64	83.17%	0.83	0.84	0.84
CNN	16, 32, 64	81.30%	0.83	0.82	0.82
CNN	128, 64, 32	80.37%	0.82	0.80	0.81
CNN	32, 64, 128	88.78%	0.90	0.89	0.89
CNN	32, 64, 128, 256	80.37%	0.83	0.82	0.82

As shown in Table V, increasing the number of neurons in the last layers has positively affected the performance. In CNNs, while simpler and lower-level features are usually obtained in the first layers, learning distinctive features is ensured in the last layers. Increasing the number of neurons in the last layers has positively affected the performance by enabling better learning of distinctive features.

Confusion matrix for the proposed model is given in Figure 4. It is seen that Other drone classes mixes with Tello, Phantom 4 and Typhoon classes to a small extent. The class names corresponding to the labels in the Confusion Matrix are given in the Table VI.

In brief, according to gathered results that are combination of different hyperparameters for CNN, final parameters employed in the CNN for classifying malicious drones are provided in Table VII.



Fig. 3. Detection and Classification Results of the Algorithm

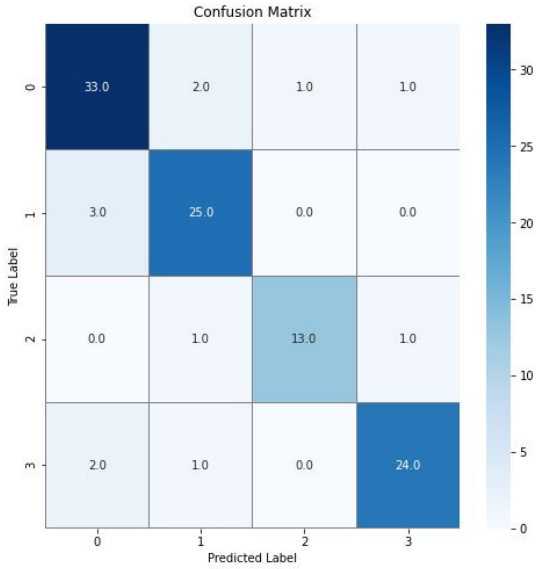


Fig. 4. Confusion Matrix Results of the Proposed Algorithm

TABLE VI. Class Label Names for Confusion Matrix

Label	Class Name
0	Other
1	Phantom 4
2	Tello
3	Yuneec

V. Conclusion

This study proposes a novel and practical methodology for detecting and classifying drones. A dataset with four classes for detecting suspicious drones was created from videos available on channels like the internet and YouTube, achieving 88.78 % classification success rate. Real-time video streams from a DJI Tello drone camera were used for classification.

In most of the studies examined in the literature,

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TABLE VII. Hyperparameters used in CNN.

Definition	Value
Input Size	200x200
Pooling Function	Average Pooling
Filter	3x3
Loss Function	Categorical Cross Entropy
Optimization	Adam
Epoch	100
Activation Function	ReLU
Number of Classes	4

the detection time has not been addressed; however, in attacks such as jamming, prolonged detection can lead to permanent damage to some electronic circuits in the drone component and make it inoperative.In the conducted study, the measurement of drone detection and classification time was aimed, and it was obtained as 0.798 seconds average.

In the scope of future studies, with the increasing the number of drone classes, after detecting a suspicious drone the intention is to seize or landing the malicious drone using various exploit software to identify its producer and communication infrastructure.

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